

TWO COUNTEREXAMPLES TO A PACKING-COLORING PROBLEM OF MORTADA, EL ZEIN, AND AL HAJJAR

ABSTRACT. Mortada, El Zein, and Al Hajjar asked whether every connected claw-free subcubic graph other than a specified graph $\mathcal{H}$ is $(1, 1, 3, 3, 4)$ -packing colorable. We give two nonisomorphic counterexamples, both cubic of order 36. Each is the triangle expansion of a cubic graph of order 12, and its noncolorability follows from an elementary analysis of the possible supports of 4-packings.

1. CONSTRUCTION AND STATEMENT

For a positive integer t , a t -*packing* in a graph is a set of vertices at pairwise distance greater than t . For $S = (s_1, \dots, s_k)$, an S -*packing coloring* is a partition $V(G) = A_1 \sqcup \dots \sqcup A_k$ in which A_i is an s_i -packing. Mortada, El Zein, and Al Hajjar asked whether every connected claw-free subcubic graph distinct from their graph $\mathcal{H}$ is $(1, 1, 3, 3, 4)$ -packing colorable [1, Problem 1].

Let C be a cubic graph. Its *triangle expansion* $T(C)$ is obtained by replacing every $u \in V(C)$ with a triangle Δ_u , whose three vertices are assigned bijectively to the three edges incident with u , and by joining the assigned vertices corresponding to each edge of C .

We define two cubic cores. The graph C_1 has vertex set

$$\{r, s, q_0, q_1, x, y, a_r, a_s, a_0, b_r, b_s, b_0\}.$$

The set $\{q_0, q_1, x, y\}$ induces a diamond with missing edge xy ; the sets

$$A = \{a_r, a_s, a_0\}, \quad B = \{b_r, b_s, b_0\}$$

induce triangles; and the remaining edges are

$$rx, ra_r, rb_r, sy, sa_s, sb_s, a_0b_0.$$

The graph C_2 has vertices p_i, q_i, x_i, y_i for $i \in \mathbb{Z}/3\mathbb{Z}$. For each i , the set

$$D_i = \{p_i, q_i, x_i, y_i\}$$

induces a diamond with missing edge x_iy_i , and the remaining edges are

$$x_iy_{i+1} \quad (i \in \mathbb{Z}/3\mathbb{Z}).$$

Put $G_1 = T(C_1)$ and $G_2 = T(C_2)$. For reference, graph6 encodings of C_1 and C_2 are K?‘CPagXAcAg and K?‘CPagPagEG.

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Theorem 1. *The graphs G_1 and G_2 are nonisomorphic connected cubic claw-free graphs of order 36, and neither is $(1, 1, 3, 3, 4)$ -packing colorable. Consequently, Problem 1 of Mortada, El Zein, and Al Hajjar has a negative answer.*

Both cores are connected and cubic of order 12. In a triangle expansion the external edges form a matching, so the replacement triangles are all the triangles. Hence $T(C)$ is connected, cubic, and claw-free, has order $3|V(C)|$, and determines C by contraction of its triangles. The core C_1 has four triangles and C_2 has six; therefore G_1 and G_2 are nonisomorphic. They have order 36, whereas $\mathcal{H}$ has order 12 [1, Figure 1]. It remains to prove noncolorability.

2. DISTANCES AND SUPPORTS

Index the vertices of Δ_u by the three edges incident with u . The vertex indexed by an edge e will be called the e -port of Δ_u .

Lemma 2 (Port criterion). *Let $u \neq v$ be vertices of a cubic graph C .*

- (1) *If $uv \in E(C)$, then every vertex of Δ_u is at distance at most 3 from every vertex of Δ_v in $T(C)$.*
- (2) *If $d_C(u, v) \geq 3$, then every vertex of Δ_u is at distance at least 5 from every vertex of Δ_v .*
- (3) *Suppose $d_C(u, v) = 2$. Chosen ports in Δ_u and Δ_v are at distance greater than 4 if and only if, for every common neighbor w of u and v , neither chosen port is indexed by the edge toward w .*

Proof. A simple core path $u = u_0 u_1 \cdots u_k = v$ lifts between chosen endpoint ports to a path of length

$$2k - 1 + \varepsilon_u + \varepsilon_v,$$

where each ε is 0 or 1 according as the chosen endpoint port is or is not indexed by the corresponding end edge of the core path. Conversely, a shortest path in $T(C)$ that uses m external edges cannot traverse one immediately in both directions, and therefore uses at least $m - 1$ intratriangle edges. Thus its length is at least $2m - 1$. The first two assertions follow. If $d_C(u, v) = 2$, every path of length at most 4 must project to a path uvw and has length $3 + \varepsilon_u + \varepsilon_v$. This gives the last assertion. $\square$

For $P \subseteq V(T(C))$, define its core support by

$$\text{supp}_C(P) = \{u \in V(C) : P \cap \Delta_u \neq \emptyset\}.$$

Corollary 3. *If P is a t -packing in $T(C)$ with $t \geq 3$, then $\text{supp}_C(P)$ is independent in C . In particular, $|P| \leq \alpha(C)$.*

Proof. The set P contains at most one vertex of each replacement triangle, and Lemma 2(1) excludes two adjacent core vertices from its support. $\square$

Lemma 4. *One has $\alpha(C_1) = \alpha(C_2) = 4$.*

Proof. Let I be independent in C_1 . If I contains neither r nor s , then it meets the diamond in at most two vertices and each of A, B in at most one. If it contains exactly one of r, s , say r , then the other vertices lie in the three cliques

$$\{q_0, q_1, y\}, \quad \{a_s, a_0\}, \quad \{b_s, b_0\},$$

so again $|I| \leq 4$. If it contains both r and s , only q_0, q_1, a_0, b_0 remain available, with edges q_0q_1 and a_0b_0 . Thus $|I| \leq 4$, and equality is attained by $\{r, s, q_0, a_0\}$.

In C_2 , an independent set meets a diamond D_i twice only in the nonedge $\{x_i, y_i\}$. Two distinct such pairs contain one of the linking edges x_iy_{i+1} , so at most one diamond is met twice. Hence $|I| \leq 2 + 1 + 1 = 4$, with equality attained by $\{p_0, p_1, x_2, y_2\}$. $\square$

3. THE TWO OBSTRUCTIONS

Call p_i, q_i the internal vertices and x_i, y_i the external vertices of the diamond D_i .

Lemma 5. *Every 4-packing in G_2 has at most three vertices.*

Proof. Suppose that a 4-packing P has four vertices and put $I = \text{supp}_{C_2}(P)$. By Corollary 3 and Lemma 4, I is independent of size four. It therefore meets one diamond, say D_i , in $\{x_i, y_i\}$ and each other diamond once. Since x_i and y_i have the two common neighbors p_i, q_i , the port criterion forces the selected ports in Δ_{x_i} and Δ_{y_i} to be the linking ports for x_iy_{i+1} and $x_{i-1}y_i$.

Independence excludes y_{i+1} and x_{i-1} . The two remaining external candidates, x_{i+1} and y_{i-1} , are adjacent. Hence one of the two remaining support vertices is internal. If it is internal in D_{i+1} , every port in its replacement triangle is at distance at most four from the selected x_iy_{i+1} -port: cross to $\Delta_{y_{i+1}}$, move within that triangle, cross to the internal vertex's triangle, and move within it. This contradicts the 4-packing condition. The case of an internal vertex in D_{i-1} is symmetric. $\square$

Suppose that G_2 has a $(1, 1, 3, 3, 4)$ -packing coloring with classes $A_1, \dots, A_5$. Each replacement triangle contains at most one vertex of each of the independent sets A_1, A_2 , and hence at least one vertex of $A_3 \cup A_4 \cup A_5$. Therefore

$$|A_3| + |A_4| + |A_5| \geq 12.$$

Corollary 3, Lemma 4, and Lemma 5 give the contradiction

$$|A_3| + |A_4| + |A_5| \leq 4 + 4 + 3 = 11.$$

Lemma 6. *Let P be a 4-packing of four vertices in G_1 , and put $I = \text{supp}_{C_1}(P)$. Then $C_1 - I$ contains a triangle.*

Proof. The support I is independent. We distinguish its intersection with $\{r, s\}$.

If $r, s \in I$, then I contains one of q_0, q_1 and one of a_0, b_0 . If $a_0 \in I$, the triangle B lies in $C_1 - I$; if $b_0 \in I$, the triangle A does.

Suppose that $r \in I$ and $s \notin I$. The other three vertices of I consist of one from each of

$$\{q_0, q_1, y\}, \quad \{a_s, a_0\}, \quad \{b_s, b_0\}.$$

If $q_j \in I$, then the three other members of I are at core distance two from r through its three distinct neighbors x, a_r, b_r . Lemma 2(3) leaves no possible port at r , a contradiction. Hence $y \in I$. The pair a_0, b_0 is adjacent. The choice a_s, b_0 leaves no possible port at a_s , because r, y, b_0 are reached at core distance two through a_r, s, a_0 , respectively; the choice a_0, b_s is symmetric. Thus

$$I = \{r, y, a_s, b_s\},$$

and $\{q_0, q_1, x\}$ is a triangle in $C_1 - I$. The case $s \in I, r \notin I$ is symmetric and leaves the triangle $\{q_0, q_1, y\}$.

Finally, suppose $r, s \notin I$. Then I contains x, y , one vertex of A , and one vertex of B . Since x, y have the two common neighbors q_0, q_1 , the port criterion forces their selected ports to be the ports for xr and ys . The chosen vertices of A and B must consequently be a_0 and b_0 , respectively, contradicting $a_0b_0 \in E(C_1)$. $\square$

Suppose that G_1 has a $(1, 1, 3, 3, 4)$ -packing coloring with classes $A_1, \dots, A_5$. As above,

$$|A_3| + |A_4| + |A_5| \geq 12.$$

Each of these three classes has size at most four, so equality holds throughout. Thus each has size four, and every replacement triangle contains exactly one vertex of $A_3 \cup A_4 \cup A_5$. Their core supports are therefore disjoint independent sets that partition $V(C_1)$. It follows that

$$C_1 - \text{supp}_{C_1}(A_5)$$

is bipartite, with the other two supports as a bipartition. This contradicts Lemma 6. Theorem 1 follows.

Remark. No minimality claim is made. By Theorem 4 of [1], both examples are $(1, 1, 3, 3, 3)$ -packing colorable. Thus they refute precisely the proposed strengthening in which the last distance parameter is increased from 3 to 4.

REFERENCES

- [1] M. Mortada, A. El Zein, and S. Al Hajjar, On $(1, 1, 2, 3)$ - and $(1, 1, 3, 3, 3)$ -packing colorings of claw-free subcubic graphs, arXiv:2608.02566v1 [math.CO], 3 August 2026.